

* RNN (Recurrent Neural Network)

Why RNN?

↳ Normal Neural Network and CNN works well When:-

↳ * input size is fixed.

↳ * data points are independent

But in real world problem are sequential.

e.g.

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Sentences - words come one after another

Stock prices - yesterday affects Today.

Speech - previous sound matters.

Video - previous frame matters.

Traditional neural Networks. cannot remember previous inputs.

RNN was designed to solve this by adding memory.

Main Concept

RNN process data step by step and remembers past information using a hidden state.

Instead of treating each input independently.

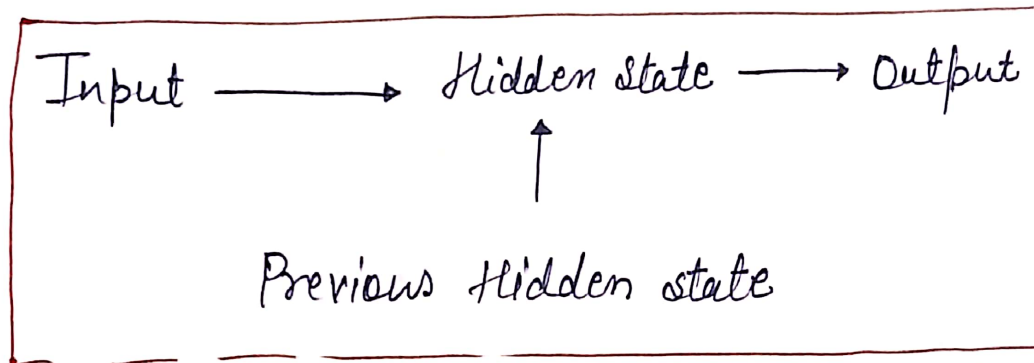
X_1, X_2, X_3, X_4 .

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RNN Says:

Current Output should depend on previous inputs too.

RNN Architecture

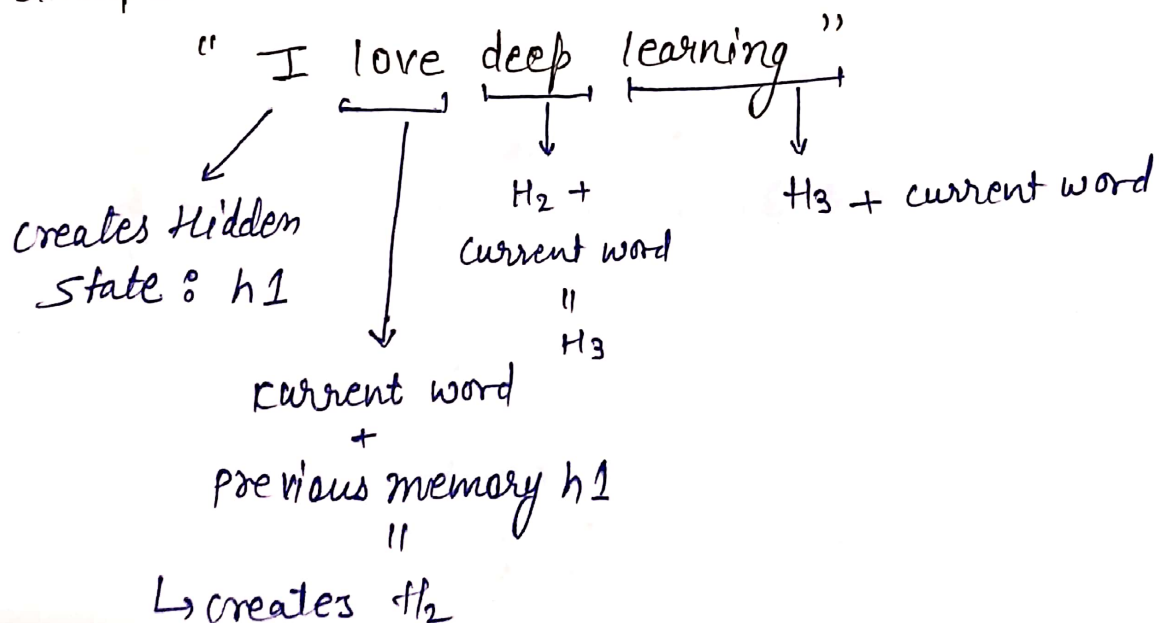


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at every timestep:

- input comes
- Previous memory is used.
- New memory is created
- output is Generated

Example:-



Important :-

* Same RNN Cell Repeats

* Same weights are reused.

This is called Parameter sharing.

Huge advantage :-

↳ fewer parameter

↳ learns patterns across sequences.

Inside an RNN cell :-

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$$h_t = f(w_h h_{t-1} + w_x x_t + b)$$

∴

activation function

x_t

current input

h_{t-1}

previous memory.

w_x

weights for input.

w_h

weights for previous state.

b

bias

h_t

New hidden state

Activation function :-

In RNN we usually us

* tanh

* Sigmoid

Output Equation

$$y_t = W y h_t$$

Hidden state converted into prediction
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Types of RNN Architecture

* One to One

(image classification)

* Many to One.

(sentiment analysis)

* One to Many

(image captioning)

* Many to Many

(machine Translation)

Training RNN :- Training RNN is special

Instead of normal backpropagation

RNN Uses :-

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Backpropagation Through Time.

Because :-

↳ same network repeat over time.